the equations. And here – surprise! – Landau was right. The gravitational field at a point is *not* well defined, when taking quanta into account.

There is an intuitive way of understanding what happens. Suppose we want to observe a very, very, very small region of space. To do this, we need to place something in this area, to mark the point that we wish to consider. Say we place a particle there. Heisenberg had understood that you can't locate a particle at a point in space for long. It soon escapes. The smaller the region in which we try to locate a particle, the greater the velocity at which it escapes. (This is Heisenberg's uncertainty principle.) If the particle escapes at great speed, it has a great deal of energy. Now let us take Einstein's theory into account. Energy makes space curve. A lot of energy means that space will curve a great deal. A *lot* of energy in a *small* region results in curving space so much that it collapses into a black hole, like a collapsing star. But if a particle plummets into a black hole, I can no longer see it. I can no longer use it as a reference point for a region of space. I can't manage to measure arbitrarily small regions of space, because if I try to do this these regions disappear inside a black hole.

This argument can be made more precise with a little mathematics. The result is general: quantum mechanics and general relativity, taken together, imply that there is a limit to the divisibility of space. Below a certain scale, nothing more is accessible. More precisely, nothing exists there.

How small is this minimal region of space? The calculation is easy: we need only to calculate the minimum size of a particle before it falls into its own black hole, and the result is straightforward. The minimum length is around:

$$L_P = \sqrt{\frac{\hbar G}{c^3}}$$

Under the sign of the square root there are the three constants of nature we have already encountered: Newton's constant *G*, discussed in Chapter 2, which sets the strength of gravity; the speed of light *c*, introduced in Chapter 3 when discussing relativity, which opens up the extended present; and Planck's constant *h*, found in Chapter 4, which determines the scale of the quantum granularity.[fn29] The presence of these three constants confirms the fact that we are looking at something which has to do with gravity (*G*), relativity (*c*) and quantum mechanics (*h*).

The length *LP*, determined in this fashion, is called the Planck length. It should be called the Bronštejn length, but such is the way of the world. In